SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA0305	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2023 / Slot C	**Date:**	13-08-2026

Code of Conduct

I _S.Mubina [192324139]_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _S.Mubina_

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2

Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.

4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

1. Analyze the semantic representations and identify the action-object relationships.

Semantic representation converts the customer's sentence into a structured form. The **action** tells what the customer wants to do, and the **object** tells which service is involved.

Customer Query	Semantic Representation	Action	Object
Activate international roaming for my number.	`ACTIVATE(Roaming, Customer)`	Activate	Roaming
Deactivate caller tune service.	`DEACTIVATE(CallerTune, Customer)`	Deactivate	Caller Tune
Check my data balance.	`QUERY(DataBalance, Customer)`	Query	Data Balance
Enable 5G service.	`ACTIVATE(5GService, Customer)`	Activate	5G Service

Analysis:

- ACTIVATE (Roaming, Customer) means the customer wants to activate roaming.
- DEACTIVATE (CallerTune, Customer) means the customer wants to deactivate caller tune.
- QUERY (DataBalance, Customer) means the customer wants to check the data balance.
- ACTIVATE (5GService, Customer) means the customer wants to activate 5G service.

Conclusion:

The semantic representations correctly identify the action and object in all four queries.

2. Determine which query contains semantic interpretation errors.

Query ID	**Actual Intent**	**Predicted Intent**	**Result**
Q1	Activate Roaming	Activate Roaming	Correct
Q2	Deactivate Caller Tune	Activate Caller Tune	**Incorrect**
Q3	Check Data Balance	Query Data Balance	Correct
Q4	Enable 5G Service	Activate 5G Service	Correct

Analysis:

Q2 contains the semantic interpretation error.

- Actual intent: Deactivate Caller Tune

- Predicted intent: Activate Caller Tune The chatbot confused

  Deactivate with Activate.

Conclusion:
Out of 4 queries, 3 are correctly interpreted and 1 is incorrectly interpreted.

3. Evaluate how meaning representation affects chatbot decision accuracy.

Meaning representation helps the chatbot understand the exact intention of the customer.

For example:

Query:
"Enable 5G service."

Representation:
ACTIVATE (5GService, Customer)

Component	**Meaning**
Action	Activate
Object	5G Service
User	Customer

When the representation is correct, the chatbot can select the correct service operation.

In Q2, the chatbot interpreted Deactivate as Activate. Therefore, the chatbot may perform the opposite action requested by the customer.

Conclusion:
Correct meaning representation improves chatbot decision accuracy. Incorrect representation can lead to wrong service actions.

4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

Improvement	**Purpose**
More training data	Helps the chatbot understand different customer sentences

Improvement	Purpose
Include opposite actions	Helps distinguish activate/deactivate and enable/disable
Use conversation context	Helps understand previous messages
Named Entity Recognition	Identifies services such as roaming, 5G and caller tune
Advanced NLP models	Improves intent and meaning understanding
Customer feedback	Helps identify and correct wrong predictions

Conclusion:

The chatbot can be improved by using more training data, context, entity recognition, advanced NLP models and customer feedback. These improvements can reduce semantic errors and provide better customer service.

2. First-Order Predicate Calculus for Smart Manufacturing.

An Industry 4.0 manufacturing plant monitors machines using semantic rules.

Production Data:

Machine	**Status**
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- Active(x) → Producing(x)
- Produces(x,y) ∧ Active(x) → Available(y)
- Maintenance(x) → ¬Producing(x) **Tasks:**

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.

4. Evaluate the effectiveness of predicate logic in industrial decision-support system A manufacturing plant has four machines.

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Given Predicate Rules

1. Active(x) → Producing(x)
2. Produces(x,y) ∧ Active(x) → Available(y)
3. Maintenance(x) → ¬Producing(x)

1. Represent the production data using First-Order Predicate Calculus

The production data can be represented as:

Active(M1)

Active(M2)

Maintenance(M3)

Active(M4)

The general rules are:

∀x [Active(x) → Producing(x)]

∀x∀y [(Produces(x,y) ∧ Active(x)) → Available(y)]

∀x [Maintenance(x) → ¬Producing(x)] Using

the first rule:

Active(M1) → Producing(M1)

Active(M2) → Producing(M2)

Active(M4) → Producing(M4) Therefore:

Producing(M1)

Producing(M2)

Producing(M4)

Using the third rule: Maintenance(M3)

→ ¬Producing(M3) Therefore:

¬Producing(M3)

Result

Machine	Status	Inference
M1	Active	Producing
M2	Active	Producing
M3	Maintenance	Not Producing
M4	Active	Producing

Conclusion:

M1, M2 and M4 are producing, while M3 is not producing because it is under

maintenance. **2. Analyze which products are currently available** The rule given is:

Produces(x,y) ∧ Active(x) → Available(y)

This means that if an active machine produces a product, that product is available.

However, the given data does **not** tell us which machine produces which product.

For example, there is no information such as:

Produces(M1,Gear)

Produces(M2,Bolt)

Therefore, we cannot identify the exact products currently available.

Information	Given?
M1 is active	Yes
M2 is active	Yes

Information	Given?
M3 is under maintenance	Yes
M4 is active	Yes
Which machine produces Gear	No
Which machine produces other products	No

Conclusion:

M1, M2 and M4 are producing, but the specific available products cannot be determined from the given data.

3. Infer whether Gear production is affected by maintenance activities

We know: Maintenance(M3)

Therefore:

¬Producing(M3)

So, M3 is not producing while it is under maintenance.

However, the question does not state that M3 produces Gear.

Therefore, we cannot conclude that Gear production is affected.

If the data had included:

Produces(M3,Gear)

then we could conclude that Gear production is affected by M3's maintenance.

Conclusion

Based only on the given information, Gear production cannot be confirmed as affected by maintenance.

4. Evaluate the effectiveness of predicate logic in industrial decision-support systems

Predicate logic is useful because it represents machine conditions and relationships clearly and allows the system to make logical conclusions.

For example:

Active(M1) → Producing(M1)

allows the system to conclude that M1 is producing.

Similarly:

Maintenance(M3) → ¬Producing(M3)

allows the system to conclude that M3 is not producing.

Advantages

Advantage	Explanation
Clear representation	Machine conditions can be represented as facts
Logical inference	New information can be derived from rules
Automation	Decisions can be made automatically
Monitoring	Machine status can be continuously analyzed
Decision support	Helps identify production and maintenance conditions

Conclusion

First-Order Predicate Calculus is effective for industrial decision-support systems because it represents facts and rules clearly and helps the system derive useful conclusions automatically.

3. **Word Sense Disambiguation in E-Commerce Search Engines.**

An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

Given Data

An e-commerce search engine receives queries containing words with more than one possible meaning.

User Query	Possible Sense 1	Possible Sense 2	Clicked Result
Apple accessories	Fruit	Technology Brand	iPhone Charger
Mouse wireless	Animal	Computer Device	Bluetooth Mouse
Java tutorial	Island	Programming Language	Coding Lessons
Python course	Snake	Programming Language	Software Development Training

1. Analyse the contextual information and determine the correct word sense for each query

The clicked result provides useful context for identifying the correct meaning of the ambiguous word.

Query	Correct Sense	Evidence	Reason
Apple accessories	Technology Brand	iPhone Charger	iPhone is a technology product
Mouse wireless	Computer Device	Bluetooth Mouse	Bluetooth Mouse is a computer device
Java tutorial	Programming Language	Coding Lessons	Coding lessons are related to programming
Python course	Programming Language	Software Development Training	Software development uses Python

Analysis

Apple accessories:

The result is an iPhone Charger, so Apple means the technology brand.

Mouse wireless:

The result is a Bluetooth Mouse, so Mouse means a computer device.

Java tutorial:

The result is Coding Lessons, so Java means the programming language.

Python course:

The result is Software Development Training, so Python means the programming language.

Conclusion

The clicked results help the search engine select the correct sense of each ambiguous word.

2. Identify the semantic cues used for disambiguation

Semantic cues are the words or information that help identify the intended meaning.

Query	Semantic Cue	Meaning Identified
Apple accessories	Accessories, iPhone Charger	Technology Brand
Mouse wireless	Wireless, Bluetooth Mouse	Computer Device
Java tutorial	Tutorial, Coding Lessons	Programming Language
Python course	Course, Software Development Training	Programming Language

Important Cues

1. **Apple + accessories** suggest a product-related meaning.
2. **iPhone Charger** confirms the technology meaning.
3. **Mouse + wireless** suggests an electronic device.
4. **Bluetooth Mouse** confirms the computer-device meaning.
5. **Java + tutorial** suggests programming.
6. **Coding Lessons** confirms programming.
7. **Python + course** suggests a programming course.
8. **Software Development Training** confirms the programming meaning.

Conclusion

The query words and clicked results provide strong contextual clues for identifying the correct word sense.

3. Evaluate the impact of incorrect sense selection on search engine performance If

the search engine selects the wrong meaning, it may show irrelevant results to the user.

Effects of incorrect WSD

- Irrelevant search results.
- Poor product recommendations.
- Lower search accuracy.
- User dissatisfaction.
- More repeated searches.
- Reduced clicks and purchases.

Conclusion

Incorrect word sense selection reduces the quality of search results and recommendations. Therefore, accurate WSD is important in e-commerce systems.

4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy

An e-commerce company can use multiple sources of information to identify the correct word sense.

Strategy	**Purpose**
Query context	Understand surrounding words
Search history	Understand user's previous interests
Click history	Identify what results the user selects
Purchase history	Understand user's product interests
Product categories	Match the query with product types
Word embeddings	Understand relationships between words
Transformer models	Understand context and meaning
User feedback	Improve future predictions

Example

If a user searches:

"Python course"

and frequently searches programming topics, the system can identify Python as a programming language.

If another user searches for snake-related products, the system may interpret Python as a snake depending on the context.

Conclusion

An industrial-scale WSD system should combine query context, user behavior, product information and modern NLP models. This will improve search relevance and recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

1. Analyze how syntactic structures contribute to semantic role assignment

Syntax shows the relationship between the words in a sentence. It helps the NLP system identify the subject, verb and object, which can then be used to assign semantic roles.

Example 1

Sentence:
"Doctor prescribed medicine to patient."

Syntactic Element	Entity	Semantic Role
Subject	Doctor	Agent
Verb	Prescribed	Action
Object	Medicine	Theme
Indirect Object	Patient	Recipient

Here, the Doctor performs the action, the Medicine is the thing being prescribed, and the Patient receives it.

Example 2

Sentence:
"Patient reported severe headache."

Syntactic Element	Entity	Semantic Role
Subject	Patient	Experiencer
Verb	Reported	Action
Object	Headache	Symptom

Conclusion

Syntactic structure helps the NLP system understand who performs an action, what is affected, and who receives or experiences it. Therefore, correct parsing is important for semantic role assignment.

2. Determine whether the assigned semantic roles are appropriate The given semantic roles should be checked against the sentences.

Entity	Given Role	Appropriate Role	Result
Doctor	Agent	Agent	Correct
Medicine	Instrument	Theme	Incorrect
Patient	Recipient	Recipient	Correct
Headache	Symptom	Symptom	Correct

Analysis

Doctor → Agent:

Correct. The doctor performs the action of prescribing.

Medicine → Instrument:

Incorrect. In "Doctor prescribed medicine to patient," medicine is the thing being prescribed. Therefore, Theme is more appropriate.

Patient → Recipient:

Correct. The patient receives the medicine.

Headache → Symptom:

Correct. A severe headache represents a symptom reported by the patient.

Correct representation

Doctor → Agent

Medicine → Theme

Patient → Recipient

Headache → Symptom

Conclusion

Three roles are appropriate, but Medicine should be Theme instead of Instrument.

3. Evaluate potential errors that could arise from incorrect parsing

Incorrect parsing can cause the NLP system to misunderstand the meaning of medical sentences.

Example

Correct sentence:

"Doctor prescribed medicine to patient." Correct

interpretation:

Doctor → Agent

Medicine → Theme

Patient → Recipient

Incorrect Interpretation	Possible Problem
Patient identified as Agent	System may think patient prescribed the medicine
Medicine identified as Agent	System may produce meaningless information
Wrong Patient-Doctor relationship	Medical record may become incorrect
Incorrect Symptom identification	Important medical information may be missed

Possible effects include:

- Incorrect medical records.
- Wrong patient information.
- Incorrect treatment information.
- Incorrect clinical decision support.
- Errors in medical information extraction.

Conclusion

Incorrect parsing can change the meaning of medical information and may lead to serious errors in healthcare applications.

4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications

The healthcare NLP system can be improved using the following methods:

Method	Purpose
Medical training datasets	Train the system using real medical language
Dependency parsing	Identify relationships between words
Semantic Role Labeling	Assign correct roles to entities
Medical NER	Identify diseases, symptoms, medicines and people
Medical dictionaries	Improve understanding of medical terms
Medical ontologies	Connect related medical concepts
Transformer-based NLP models	Improve contextual understanding
Rule-based validation	Detect possible semantic errors
Expert review	Verify important medical information

Conclusion

Healthcare NLP systems should combine accurate syntactic parsing, Semantic Role Labeling, medical Named Entity Recognition and domain-specific NLP models. These methods can improve the accuracy and reliability of medical information extraction.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
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